

Predicting Cybersickness Trend and Extent based on FMS labeled Dataset

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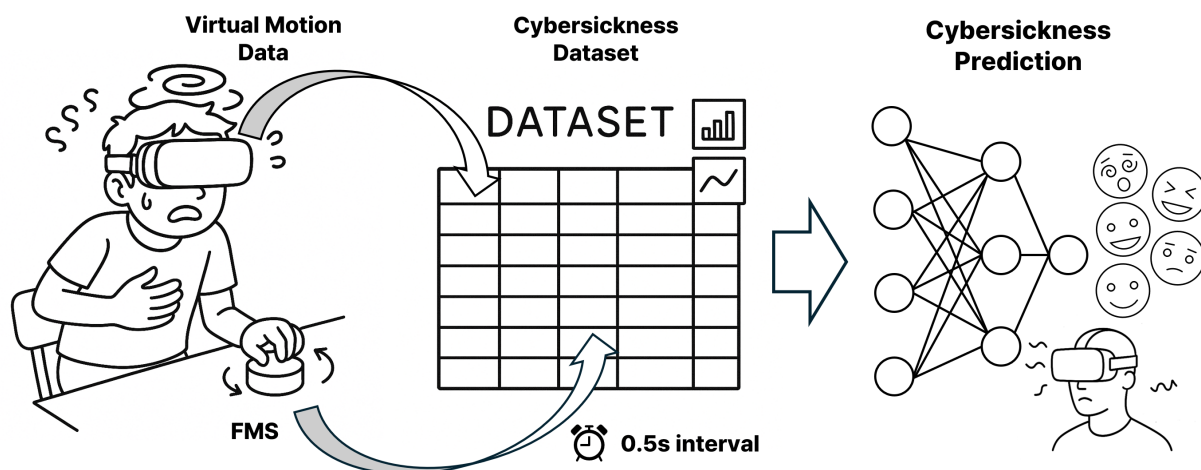


Fig. 1: A temporally dense dataset (over 179,340 points of cybersickness data) is developed. The FMS score was marked every 0.5 seconds and adjusted/updated on-demand, and the dataset is publicly available. Predicting cybersickness with the dataset shows the possibility of predicting cybersickness with only virtual motion profiles, which are comparatively easier to obtain.

Abstract—Cybersickness has been a hindrance to the widespread adoption of virtual reality. As cybersickness is dynamic, with its manifestation changing constantly, predicting and mitigating it in real time is key. However, previous research has utilized predictive models trained mostly with datasets whose sickness levels were measured and labeled only after experiencing the content. In addition, many datasets rely on physiological signals as input, which makes them difficult to apply in actual VR usage. This makes such timely predictions unreliable or practically infeasible. We have created a publicly available dataset where the ground truth sickness levels were densely marked every 0.5 seconds and adjusted/updated on-demand using the FMS, and purposely excluded the difficult-to-collect physiological data. We demonstrate that predictive models trained with such a dataset, comprising just the content motion profile and FMS data, can still produce comparably reliable sickness prediction, and more so, when user-specific parameters (e.g., age, gender) are added. The created dataset is available at https://github.com/pvtryan1024/2025_Cybersickness_dataset.

Index Terms—Cybersickness, Datasets, Virtual reality, User studies, Real-time sickness prediction

1 INTRODUCTION

Cybersickness, which has been reported to occur in more than 60% of VR users, has been a persistent problem since its wider adoption [40]. Although multiple theories have been proposed as the main cause of cybersickness, the most prevalent hypotheses are the sensory mismatch [32], postural instability [2], and rest frame theories [8]. In one way or another, the virtual (or visual) motion seems to be an important factor in all these theories. For example, sensory mismatch emanates from the discrepancy between visual and vestibular motion information. There have also been reports of patients with vestibular migraine showing postural instability of the head and neck when visual stimuli were given in their peripheral view [28].

While the induced perception of self-motion (vection) might not be

a definitive cause of visually induced motion sickness (VIMS) [20], it is likely to show high correlation with cybersickness. As such, a dataset of users' virtual motion associated with the level of cybersickness could be useful in studying the nature of cybersickness and even predicting it. In fact, several previous studies have collected and used Multimodal Time-series Sensor Data [13] in hopes of finding a correlation with a predictive model of cybersickness. These datasets typically capture, in addition to the virtual motion profile, various types of data such as physiological signals [43] (e.g., heart rate, gaze). Predictive models based collectively on such data can be difficult to deploy in real usage situations, during which physiological input data is practically difficult to acquire. Furthermore, it has not been clearly established if or how much the various physiological signals actually correlate with cybersickness [9, 13].

The most serious problem seems to be that the datasets to this end use ground truth cybersickness levels measured and labeled after experiencing the content, most commonly the Simulation Sickness Questionnaire score [16]. However, cybersickness is dynamic by nature and is constantly affected by various factors, moment by moment. This makes such analysis and predictions unreliable, as we are unable to consider

various factors that appear and disappear at different times.

In this paper, we have created a publicly available dataset containing the user's (or content) virtual motion profile (e.g., angular velocity, acceleration) in which the ground truth sickness level was densely marked every 0.5 seconds, and updated/adjusted on demand using the FMS [35], using a dedicated, easy-to-use dial based input device [30]. In addition to making the data set temporally dense, the resolution of 0.5 seconds reflects a conservative estimate of how fast a human user can update the FMS value using the dial. In addition, we purposely do not use or collect any physiological (or other type of) data (to be used as input to the prediction model) that are difficult to collect in a usual VR setup.

We demonstrate that using such a dataset and training a prediction model with it can produce "timely" sickness prediction performance comparable to those that use, for example, physiological data. Note that models based on a dataset with single SSQ score-like labels can only produce an "overall" or average predictive extent of the sickness and fall short of making moment-to-moment predictions. This is especially problematic for highly interactive content. We also observed that adding personal traits can further improve the prediction performance and reflect the individual differences. Our contributions are summarized as follows:

- **An ecologically valid and densely collected time series cybersickness dataset:** A time series cybersickness dataset is compiled, reporting the FMS score densely every 0.5 second with more than 150 users (more than 400 data points for each participant) and for a variety of virtual motion types (e.g., all six degrees of motion). All data collection sessions lasted more than 5 minutes, unlike other studies [22, 25, 36], where each session took less than 2 minutes or even 90 seconds, which might not be long enough to experience cybersickness. In addition, it was ensured that the carry-over effect [46] was minimized by strictly adhering to the one-day-one-experiment policy with 16 hours between the collection sessions. The link for the dataset is provided on [this GitHub page](#), identical to the link on the abstract.
- **On-line prediction of the cybersickness trend and extent based on the dataset:** We demonstrate the feasibility and power of the "moment-by-moment" cybersickness prediction model trained with the aforementioned dataset. We compare its performance to that of the model trained on a dataset that only has temporally sparse measurements (the extreme case being just a single SSQ-like label per session).
- **Ablation study that illuminates the comparative effects of different factors on cybersickness:** Through the ablation study, different factors, objective and user-specific, are compared in terms of their effects on cybersickness. In particular, we also show that the characteristics of virtual motion, which are mainly agnostic to individual differences, are correlated with the behavior of cybersickness. This is made possible by relating the ever-changing motion profile to the moment-by-moment sickness labels. It validates the long-believed hypothesis that motion characteristics are a significantly important objective feature for sickness assessment.
- **Demonstration of using user-specific data to model individual differences:** By incorporating static user-specific data (e.g., age, gender, MSSQ) into the basic virtual motion data, we demonstrate how prediction accuracy can be improved and tailored to individual users.

2 RELATED WORK

2.1 Virtual Movement and Cybersickness

Visually induced motion sickness has been a persistent problem when using virtual reality systems. Typical symptoms include eye strain, headache, nausea, and in severe cases vomiting [24]. Attempts to explain cybersickness with visual motion and vestibular stimulus have been made in multiple works. Bos et al. [3] created a hypothetical

model that considered the visual flow and subjective vertical axis to explain cybersickness. The perception of the vertical axis via vision or from the otoliths within the vestibular organ has effects on the severity of cybersickness.

Although the effects of visual motion stimulus and its effects on cybersickness are inconclusive [19, 20], it does seem to have some correlation when it comes to the axis of rotation. Ujike et al. [38] have tested all three axes of rotation with a scale of severity of cybersickness from 0 to 10. The experiment results showed that the rotation around the roll axis incurred the highest sickness score, followed by pitch and yaw.

However, more recent experiments by Keshavarz et al. [19] have found that while there was not a huge difference among the rotational axes, the pitch rotation showed generally higher Fast motion sickness scale and SSQ scores. Note that rotation usually occurs around multiple axes simultaneously. Keshavarz et al. [17] show that visual rotation with more than two axes will induce higher cybersickness than a single-axis rotation. Another factor that should be considered in visually induced motion sickness is the frequency of the movement.

While the "precise" relation between cybersickness and virtual motion profiles is still under investigation, there are multiple works that at least point to the close mutual correlation. Thus, in this work, we opt to use only, for practicality's sake, the virtual motion profile (to predict or assess the basic extent of cybersickness) and, optionally, personal characteristics information (to address individual differences) using the machine learning approach.

2.2 Cybersickness Datasets

2.2.1 With physiological signal data

Several datasets have been created by researchers to develop cybersickness prediction models, particularly using various psychological data. Jeong et al. [14] collected EEG and cybersickness data while participants viewed 360° videos, using a keyboard to assess cybersickness by pressing a key whenever they felt sick. There was a 3-minute rest period between the 360° video viewings, but this is too short to recover from the induced cybersickness symptoms; therefore, the reliability is questionable [46].

Smyth et al. [36] collected cybersickness data using the FMS method [18] with the EDA (Electrodermal Activity) and skin temperature data, but could not establish any significant inter-relationship. Xue et al. [44] collected emotion data and physiological responses (also EDA and skin temperature) with SSQ scores to create an emotion-related dataset for 360° videos. However, the study was mostly focused on the affective state (e.g., valence and arousal) and its correlation with the physiological data.

Kim et al. [21] created a 360° video dataset with variable frame rate and physiological signals and SSQ responses. The sickness score was collected after viewing each video, lasting 60 seconds each. Dong et al. [6] created a dataset with head movement, heart rate, and a cybersickness survey while playing multiple virtual reality games. The SSQ was collected after playing a game for 3 to 5 minutes.

While cybersickness prediction or assessment models based on such datasets have shown varying degrees of success, collecting physiological data is either impractical or costly in a real-life VR usage scenario. Most VR devices do not include physiological sensors (e.g., EEG, HR, EDA, ECG, etc.). Aside from requiring additional external devices, which most users do not have, there are conflicting reports on whether physiological signals do, in fact, correlate with cybersickness. For example, a work from Nalivaiko et al. [31] reports that heart rate has correlation with cybersickness, while Dennison et al. [5] did not find such effects.

2.2.2 Temporal resolution

Most cybersickness related datasets use post-evaluation — that is, sickness assessment is done (e.g., using the SSQ [16]) after viewing the given test content. This is partly due to the large number of questions to be answered, which is not feasible during the test. For this, in-situ on-line assessment methods such as the FMS have been suggested [18], in which the user is asked to make a very simple and short assessment

of the sickness while experiencing the test content. This way, sickness data can be collected, at least partially, at a denser temporal resolution.

In the work of Lee et al. [25], SSQ scores were collected every 90 seconds from 20 subjects. The image frame itself and EEG, EKG, and GSR were also acquired. However, as mentioned above, as there are 14 SSQ questions to be answered, the lengthy interruptions (to answer the questions) could have affected the data, possibly in an unknown way.

Kim et al. [23] collected a subjective measure of cybersickness from 1 to 5 (simple), with 44 VR contents per participant. Each VR content only lasted 30 seconds, which does not seem long enough to induce a sufficient extent of cybersickness. Moreover, the test's total duration of over 2 hours (44 different 30-second VR contents) was excessive, and the 3-minute resting period between tests was relatively insufficient for recovery from the ensuing sickness.

The work from Islam et al. [11] collected FMS every 30 seconds during the session and SSQ at the end of every session. While this follows the very purpose of FMS, i.e., on-line sickness data collection, data with the 30-second collection period may be too sparse to be a baseline for a good prediction/assessment model. From a more recent work from Jeong et al. [13], the experiment consisted of four 45-second videos for each session, and took FMS every 15 seconds.

Similarly, Wen et al. [43] collected the cybersickness response every 30 seconds by voice. While the dataset includes a rich set of support data, such as the head position, gaze, content motion vector, and physiological signals, such as heart rate, the 30-second sampling period is likewise deemed too long. In addition, at the time of this writing, it is not publicly accessible.

In summary, for many of the major datasets we have reviewed, the test duration was either too short to induce a sufficient level of cybersickness in a single session, being less than a minute in extreme cases, or the data was collected too sparsely, ranging from 15 seconds to 3 minutes. It is quite clear that a higher temporal resolution would be needed for effective real-time sickness prediction — e.g., sickness levels can change significantly during 15 seconds. In addition, the overall length for each participant was too long with minimal rest time, which would have definitely caused the carry-over of cybersickness as the test progressed. The unavailability of the dataset defeats the very purpose of a dataset (like cross-comparison and avoidance of redundant work) and further brings about doubts about its integrity. This work provides a public dataset with the ground truth sickness data collected using the conventional on-line method (such as the FMS) densely (probably the densest to date), with data present at every 0.5 seconds. The resolution of 0.5 seconds would be more than sufficient for moment-by-moment sickness prediction.

2.3 Cybersickness Prediction

Predicting cybersickness using physiological or visual data has been tried on many occasions. However, recent cybersickness machine learning approaches have shown some troublesome trends of either the dataset not being public or the authors neglecting to provide the appropriate performance metrics. Before the (highly effective) deep learning techniques became popular for classification/prediction tasks, optimization and regression have often been used for finding and analyzing the correlation between data variables (e.g., cybersickness scores and the probable factors).

So et al. [37] used the notion of spatial velocity (SV) to predict vection in a given scene. Linear regression was used to find the high correlation between SV and SSQ [16], e.g., compared to the visual fore-and-aft movement or with the yaw axis rotation. Padmanaban et al. [33] used the decision tree in their attempt to relate SSQ scores to their own handcrafted features that captured the visual change in the content. While the error was quite high compared to the more recent works, it showed a departure from the more classical linear regression method and could capture non-linear relations between the proposed features and cybersickness.

Liao et al. [27] used the DNN (LSTM and dense layers) with the brain wave (EEG) data to predict cybersickness. The resulting performance with MAE (mean absolute error) of 0.1 (in the best case) is

somewhat dubious, as the cut used for motion sickness determination was not provided.

Weina et al. [15] used both the CNN and LSTM-RNN architectures to predict cybersickness. The best case result showed an MSE of 0.009, which is rather questionable since the R^2 was only 0.868 when the MSE indicated a near-perfect prediction from the ground truth dataset, labeled in 0 to 11-point scale.

Jeong et al. [14] used CNN networks, using the spectrogram of the EEG signals as the input. While the accuracy was above 90%, the paper did not provide other important metrics such as precision or recall. Therefore, it is difficult to assess its true performance.

Lee et al. [25] used a variety of DNN techniques (LSTM, 1/2/3D CNNs) to extract various features from the content and physiological responses, and attempted to regress and predict the SSQ score. The RMSE, PLCC [7], and SROCC [1] metrics have shown reasonable performance, however the dataset is unavailable. Kim et al. [23] used a similar approach to distinguish among cognitive, visual, spatial, and temporal features from their dataset (which is not publicly available) and predict the extent of cybersickness in five levels. The confusion matrix of the class prediction results was provided, and it was one of the few examples where the metrics were provided in detail, but the dataset is not available.

A recent work from Islam et al. [11] tested different DNN architectures for sickness prediction (including the CNN-LSTM models, Deep Temporal Convolution Networks with 1D convolutional layers and residual networks, and attention-based LSTM) with their own dataset, which collected FMS data every 30 seconds. While the input range of the FMS scores was between 0 and 4, the small MAE and RMSE (0.5 to 0.6) and sMAPE being just 40%, conversely, hints that the ground truth FMS score might have ranged only between 0 and 2 (i.e., not sufficiently high sickness induced in the first place) or scale normalization might have been needed. Confirmation is difficult due to the unavailability of the dataset.

Wang et al. [42] used LSTM and hidden layers for a binary classification task, but there was no mention of recall, precision, or other metrics. The dataset is not publicly available.

Another work from Jeong et al. [13] utilized transformers with specified encoders, providing MAE and RMSE, but it also shares the same issue of not clarifying the ground truth data range; consequently, it is unclear if the prediction is sufficiently plausible. Comparison of MAE and RMSE with other works is also provided, but the datasets of the compared works have different scales.

Lee et al. [26] used the SSQ data from Padmanaban et al. [33] and augmented the dataset using the videos mentioned in Padmanaban's work. 3D convolution networks were used and yielded better results compared to Padmanaban's work, showing a higher correlation in predicting the cybersickness of some videos. Although the Pearson linear correlation coefficient (PLCC) appears promising, the table from the work reveals a 14.9 difference in predicting the SSQ score for a specific video. The prediction was also made on the SSQ score of each video and not on specific points of the experience.

Thus, while applying machine learning and DNN to cybersickness seems to be the natural and obvious choice, to this end, the overview of previous work illustrates that it is still not clear what kind of data needs to be collected — most datasets simply try to include all probable features such as the content itself, virtual motion information, physiological signals, and user specific traits. This, in turn, also makes it difficult to converge on the proper DNN architecture. The difficulty in leveraging previous work also stems from the failure to report comparative performance statistics in a proper manner. As also pointed out in the previous subsection, most prediction models have not used datasets with sufficiently temporally dense sickness valuation, albeit simple — perhaps the best case being the dataset with 30-second period FMS data [11]. The Table 7 in the Appendix summarizes the features of some of the major cybersickness datasets and prediction models.

3 DATASET COLLECTION

3.1 Collection Procedure

3.1.1 Data Collection

The dataset was collected and accumulated over four different cybersickness related experiments. While these experiments were designed differently for various objectives (and different subject pools), the data were collected in the same unbiased way at the individual session level — i.e., using the same test content and procedural logistics.

The first experiment was a between-subjects design, thus there was no need for consideration of the rest intervals and carry-over effects. The remaining three experiments were all within-subjects, and for these cases, the participants had at least a day between the experiments to acquire more trustworthy cybersickness data without the carry-over effect [46]. For lack of space, we omit the detailed description of all the experiments, but as already indicated, all the individual sessions (or treatments) involved the same test content and procedural and data collection logistics. By such a measure, the integrity and bias-free nature of the dataset is preserved.

3.1.2 Test Session

The individual test session, during which data collection took place, began with a pre-experiment survey. The participant was informed about the experiment and notified that the test could cause mild to serious cybersickness. A consent form was filled out along with personal and background information. The test content, which consisted of a virtual space flight (developed using Unity game engine [39]), lasted around 4 minutes for each session, and depending on the experiment, the participants experienced up to five repetitions of the virtual environment.



(a) Sickness scale (icons) from "No" sickness to "Severe" sickness



(b) The sickness scale shown to the user during and within the test environment.

Fig. 2: The sickness scale (and the corresponding facial icon) and the current value shown to the user during the test whenever the FMS input is made, and its value is changed. The icon and scale would fade within a second when no inputs are made to minimize the rest frame effect [34].

3.1.3 Test Content and Set-up

Meta Quest 3 headset (resolution of $2 \times 2064 \times 2208$ and 90 to 120Hz refresh rate) [29] was used for all data collection sessions. A high-end PC (AMD Ryzen 7 5800X, NVIDIA RTX 3070Ti, 64GB RAM, running Windows 11) was used to run the test content, maintaining a frame rate of higher than 72 Hz in all cases. The motion profile data of the test content (see Figure.3) was collected within the content through the Unity engine — that is, the velocity, acceleration, and angular velocity calculated from the properties of the virtual player's headset object.

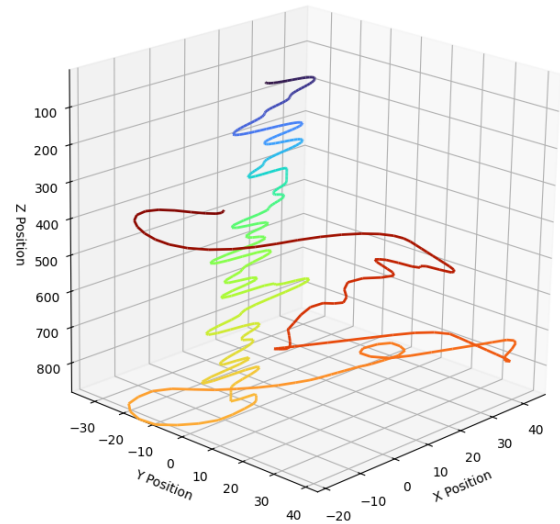


Fig. 3: Trajectory (X,Y,Z convention) of the virtual user movement within the sickness-inducing test environment

The FMS score, used as the ground truth data for cybersickness, was recorded every 0.5 seconds in-game time using an external device (Microsoft Surface Dial). The dial device was chosen for making relative input fast — namely, adjusting the current score on demand. A slider [4] or a dedicated key/button device [45] would also have sufficed. Direct (absolute) input method was avoided as the time needed for the user to think and identify the right input key or value would interfere with the task and produce bias in the data. That is, once the score was initialized and set at the beginning, the current value was shown to the user, then the user used the dial to make it higher or lower — if not, the score was kept the same and recorded as such. Adjusting the dial functions would change the FMS value as shown in Figure.4b, increasing the value when turned clockwise and decreasing the value when turned the other way. The motion profile data was saved at the same rate as the FMS score. See Figure4.



(a) Test environment.



(b) The Microsoft Surface dial

Fig. 4: FMS was collected in real time during the experiment. Note the Surface dial held on the right hand. Upon activation, the current score was shortly displayed, and the FMS scores were adjusted by turning the surface dial.

3.1.4 Institutional Review Board Approval and Participant Data Handling

We have received ethical approvals for all the experiments within which the data collection sessions were run from the Institutional Review Board of Korea University, and all participants' data were anonymized in the dataset. All participants were allowed to withdraw from the experiment if they wished to do so. A precaution explaining the side effects of the experiments was given before signing the consent form. The collected personal information will be deleted after the required storage period required by the Institutional Review Board.

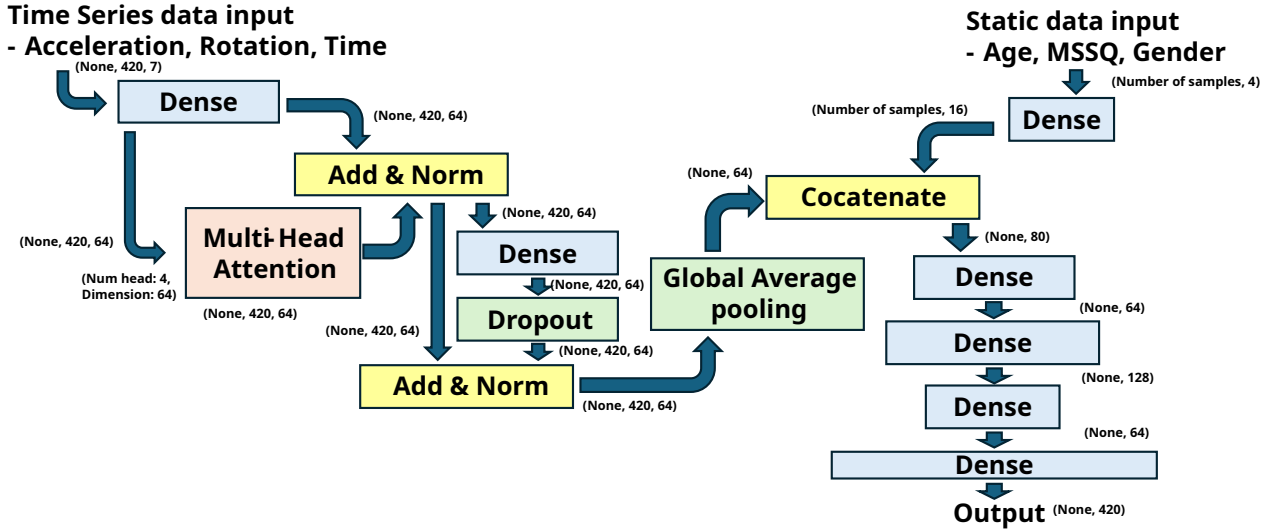


Fig. 5: Multihead Attention-based cybersickness prediction model.

3.1.5 Participants

For the first experiment, 48 subjects were recruited from the online community of our institution. The participants consisted of 18 males and 30 females with an average age of 25.52 and a standard deviation of 3.31. The dataset was collected during April 2024.

In the second experiment, 37 subjects were recruited from the same online community of our institution. The participants consisted of 13 males and 24 females with an average age of 24.58 and a standard deviation of 4.06. The dataset was collected during September 2024. The experiment was a within-subjects experiment on cybersickness with 3 conditions for each participant.

In the third experiment, 31 subjects were recruited from the same online community of our institution. The participants consisted of 19 males and 12 females with an average age of 24.27 and a standard deviation of 4.33. The dataset was collected during October 2024. The experiment was a within-subjects experiment on cybersickness with 4 conditions for each participant.

In the fourth and last experiment, 40 subjects were recruited from the aforementioned online community of our institution. The participants consisted of 20 males and 20 females with an average age of 26.15 and a standard deviation of 4.40. The dataset was collected from February to April 2025. The experiment was a within-subjects experiment on cybersickness with 5 conditions for each participant.

4 DATASET OVERVIEW

Table 1: Summary of the collected cybersickness related data.

Data Source	Collected Features
Unity Engine	Timestamp
	X axis Acceleration
	Y axis Acceleration
	Z axis Acceleration
	X axis angular velocity
	Y axis angular velocity
	Z axis angular velocity
External Device	FMS
Off-line Survey	Age
	MSSQ
	Gender

The data distribution for the dataset is illustrated in on Figure.7 located on the Appendices section. The collected features can be found on Table 1. For variables that were collected in real time, FMS score

had a mean of 6.37, a standard deviation of 5.37, X axis acceleration had a mean of -0.0014, a standard deviation of 4.45, Y axis acceleration had a mean of -0.024, a standard deviation of 5.14, Z axis acceleration had a mean of -0.045, a standard deviation of 3.13, X axis angular velocity had a mean of -1.44, standard deviation of 20.80, Y axis angular velocity had a mean of -0.046, standard deviation of 31.32, Z axis angular velocity had a mean of 4.85, standard deviation of 34.06. For the static features, the MSSQ score had a mean of 15.45, a standard deviation of 8.74, and age had a mean of 24.90, a standard deviation of 4.24. There were 246 experiments conducted on females and 181 on males.

5 DATA UTILIZATION

5.1 FMS Score Prediction

We tried out four prediction models to handle and capture the cumulative effects of the time series cybersickness data, including a basic linear regression model, a paired 1D CNN [42], an LSTM [11], and a transformer [13] (multi-head attention model [41]). The details of the respective architecture and associated parameters are outlined in Table 2.

In particular, the details of the multi-head attention model are illustrated in figure5. Similar to the structure shown in the figure, static features such as age and MSSQ scores were merged with the processed time series data that went through the multi-head attention layers. The gender feature was one-hot encoded, and all static features were standardized. The acceleration data had some outliers, in which both clipped values and original values were used to train and validate the network. The result did not differ; therefore, the original values were used. The data input for all four models was the same and was pre-processed before use. Since data collection experiments differed in their duration (and number of data points), all were cut at the 420th data point per experiment to ensure that the input vectors are usable for the models. Experiments with less than 420 ground truth FMS data were discarded (but are included in the dataset). The collected data were used to train the model for 30 epochs for all models.

5.2 FMS Score Prediction Results

Table 3 shows the respective performances of the four prediction models. In addition to RMSE, MAE, and R^2 , to measure the accuracy regardless of the scale of the ground truth data, we also report on the sMAPE.

The linear regression coefficients showed that the personal features, such as age (0.1595) and MSSQ score (0.1229), had a high correlation with the FMS score. The rest had values of less than 0.05. The results

Table 2: Summary of the three DNN architectures for time-series + static input modeling. Static input contains data features that were collected by the survey (e.g. personal data).

Model	Module	Layer	Parameters / Notes
CNN-based	Time-series encoder	$3 \times \text{Conv1D}$ (64, 128, 128 filters, kernel size=25, padding=same) + BatchNorm + Dropout(0.3)	ReLU activation, temporal features extracted
	Static encoder	RepeatVector(420) + Dense(64, ReLU)	Static features projected and aligned to match time series data sequence length
LSTM-based	Time-series encoder	LSTM(128, return_sequences=True) + LSTM(64, return_sequences=True) + BatchNorm + Dropout(0.2)	Sequence-to-sequence modeling, tanh activation (default)
	Static encoder	RepeatVector(420) + Dense(64, ReLU)	Static features projected and aligned to match time series data sequence length
Transformer-based	Time-series encoder	Dense projection $\rightarrow$ Multi-Head Attention(16, 64) + Residual + LayerNorm + Feedforward (Dense(64, ReLU) + Dropout)	Temporal encoder with residual connections, dropout(0.2)
	Static encoder	Dense(16, ReLU) + fusion after GlobalAveragePooling1D	Static input combined after processing time series data

show that the multi-head attention model generally performed the best in terms of RMSE, MAE, R^2 , and sMAPE.

Table 3: Performance measures between the four prediction models.

Models	RMSE	MAE	R^2	sMAPE
Linear Regression	4.2853	3.3632	0.3699	74.64%
1D CNN + 1D CNN	4.2706	3.3704	0.2675	78.52%
LSTM	4.1925	3.3044	0.2940	75.79%
Multi-head Attention*	3.8133	2.9831	0.4159	73.05%

5.3 Comparison to Prediction with Sparse Dataset

As existing datasets, many of them not even available, all have different data formats and time resolutions (see Table 7), it is difficult to compare the performance of prediction models between using a dense dataset like ours to a particular sparser one. In fact, it is almost obvious that the latter could not perform accurately for making predictions or assessments at a high rate.

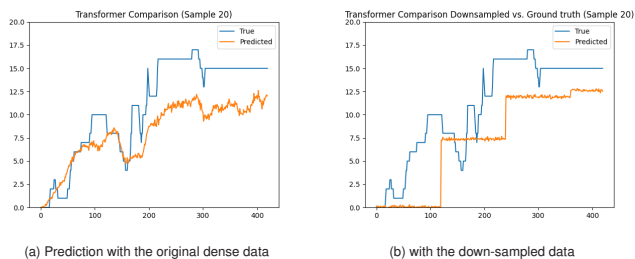


Fig. 6: The prediction made with the down-sampled ground truth data fails to capture the downward changes in cybersickness between the data sampling period (blue: ground truth, orange: predicted)

Nevertheless, to mimic the effects of a longer interval for collecting cybersickness, we have down-sampled the FMS data to have the same value for every 120 data points, which corresponds to 60 seconds in real time, which is the most common interval that is used (in the previous work) to collect FMS scores, to our knowledge. The results in Table 4 seems to indicate that the overall accuracy of the predictions has

Table 4: Down-sampled data against real-time data. Note that even though the prediction results seem to improve when the ground truth samples have a lower resolution, it actually performs worse considering the actual ground truth data.

Used data	RMSE	MAE	R^2	sMAPE
Low res prediction	3.5631	2.4530	0.5229	98.09%
Against Ground Truth	4.2393	3.1813	0.2781	91.93%
Ours	3.8133	2.9831	0.4159	73.05%

increased when both the predictions and the results are derived from the down-sampled data (with the exception of sMAPE). However, in reality, when compared with the actual ground truth FMS data collected every 0.5 seconds, it shows that the prediction results are actually worse, failing to capture the moments where the cybersickness actually decreases from time to time. An example of such a performance sample is shown in Figure 6. From the figure, it can be seen that the prediction may be somewhat correct only at a certain period matching that of the data samples.

5.4 Ablation Study

Table 5: Performance of the transformer with feature removed

Removed Feature	RMSE	MAE	R^2	sMAPE
Personal Data	4.0687	3.220	0.3351	75.75%
Acceleration	3.8338	2.9905	0.4096	73.27%
Angular Velocity	3.8093	2.8979	0.4171	73.39%
Timestamp	3.7109	2.8978	0.4469	72.00%
All Included	3.8133	2.9831	0.4159	73.05%

Table 5 shows the results of the ablation study on the best-performing transformer model, isolating certain input data.

The error increased when personal information-related features were removed from the model. This could indicate that personal features do have some impact when it comes to predicting cybersickness.

The model performed better when angular velocity was removed; however, considering that the sMAPE score became higher, it is questionable whether this was an actual improvement over the base model.

The model also performed better without the time stamp, suggesting that the duration of the experience might be less impactful than the movement within the VR environment or the individual differences and characteristics of the participant.

Table 6: Dynamic time warping value difference with and without the static features (namely - Age, Gender, MSSQ) for three participants.

Participant No	DTW value with static feature	DTW value without static feature
60	268.24	989.41
90	297.28	437.43
100	535.29	984.10

While the prediction results from the multihead attention gave around 3 MAE, they indicated that the trained model could capture some of the individual differences among the subjects, as shown in Table 6. Dynamic time warping can capture the similarity between the time series data [12]; therefore, we used it as a metric to show an example of how similar the patterns are between the ground truth cybersickness. The analysis was made on each individual experiment. The total sum of the dynamic time warping (DTW) value also decreased when calculated for all subjects. Not all subjects showed a decrease in their DTW values; however, any increase was minimal. The predicted value changed significantly in magnitude according to the age or gender and the MSSQ score of the participant.

6 DISCUSSION

Currently, there are no accepted standards on the set of features that are to be collected as a basis for predicting cybersickness. This has prompted various researchers to work on differently composed datasets. As such, a bundle of likely features has been included, such as the content imagery and motion data, physiological signals (e.g., heart rate, EEG, ECG, GSR, gaze), personal information, symptom survey, head movement and balance data, making the collection process possibly inefficient.

Furthermore, it is important to note that much care should be taken in how the data are collected. The test contents should specify guidelines on the threshold of how long one should be exposed to elicit sufficient cybersickness to extract meaningful data, at least for an average person, and also the time interval for rest between the trials. Ideally, to be conservative, as previous literature suggests, the rest period between the sessions should be at least a day or more, not just for data integrity and preventing the carry-over effect [46], but also for the safety of the participant.

The method of collecting the ground truth for cybersickness should also be discussed, especially in light of the objective — e.g., whether it is for on-line prediction or an overall prediction for the whole experience. Symptom detailed dense on-line data are difficult to collect due to their variety. This work collected the ground truth for cybersickness based on the work of Keshavarz et al. [18] — it did not report the symptoms of cybersickness, but just a subjective rating of the extent of cybersickness. Many studies collect simplistic data in a binary fashion — sick or not — without measuring the extent of the symptom, making it difficult to make an effective prediction. Finally, to make on-line prediction (and apply mitigation methods right away), the sampling period of the sickness information or extent should be as short as possible.

7 LIMITATION AND FUTURE WORK

The ground truth data for the dataset in this paper were collected using the dial device in real time. However, by the nature of the dialing, for data entry of rapidly changing values, the process can be time-consuming and cumbersome — e.g., having to dial through 2 to 8 to go from 1 to 9 (sudden occurrence of sickness).

Compared with the previous works mentioned in Section 2, our prediction results show lower accuracy. We believe this is due to the high individual differences between the users. Our goal was to provide a dataset and show that cybersickness could be predicted well enough

without physiological data. Better and refined machine learning models might improve the prediction results. This remains a task for the future, and those who are interested can utilize our dataset to create a model for predicting cybersickness.

The entire data was collected on the same test environment; therefore, the motion profile data values that were used to train the networks were, in effect, the same for every individual. Even so, the FMS data tended to differ widely among users, which could be due to the inherent individual differences, but also due to the fact that the FMS method itself cannot escape from being quite a subjective measure as well. We have chosen a space environment to include all 6 axis of motion (pitch, yaw, roll rotation, X, Y, Z axis movement), and were able to capture the feasible motions possible within the virtual reality environments. Another factor in choosing a space environment was that it was abstract enough to be generalizable over environments such as roller coasters or racing, etc. However, it does not change the fact that a single motion profile was used for the dataset. We plan to create a heuristic algorithm that creates a trajectory for each experiment so that it can cover all ranges of possible motions within a virtual environment while maintaining its overall time per experiment session.

The data was marked at a 0.5-second interval, which was deemed sufficiently continuous and a value during which a significant change in sickness would not occur. In addition (according to our rough measurement), a human user would be able to change up to 8 level values in a single rotation within 0.5 seconds, thus it would still support even a rare case of inputting fast-changing values. However, the optimal collection interval remains to be a topic for further research.

Our dataset was created using participants mostly in their mid-twenties, with little variance. This has limited the dataset in terms of age distribution, which might limit its capabilities in capturing cybersickness trends on different age groups.

For our future work, we plan to address the issues mentioned above while investigating more information and features that can be collected using a typical VR setup to determine their relative importance to cybersickness. One could also surely envision that sensors for certain physiological signals (if proven to be a significant contributing factor to cybersickness) may become a standard component in future VR headsets or controllers (e.g., like eye trackers have recently). Our data set was created using only one test content. Although it included a variety of motion profiles, it would be desirable to collect more data with a more varied content and motion combinations.

8 CONCLUSION

In this work, we have created a public dataset, useful for building an on-line cybersickness prediction model. The most critical part of the dataset is that the collected data has high temporal density, making it most proper to train on-line prediction models. In addition, while most cybersickness datasets have been collected in a way that is vulnerable to the carry-over effect, our dataset was collected by strictly complying with the one-day-one-experiment policy, minimizing the risk of collecting unreliable cybersickness data. It has been collected through more than 140 subjects with 420 cybersickness measurements each, giving more than 179,340 datapoints for this dataset without including the additional feature vectors.

Even with a very simple DNN model, training with the given dataset yields reasonable prediction performance. Moreover, it shows superior performance, albeit somewhat obvious, even without the often-used but difficult-to-apply-in-practice physiological signals, compared to training with a sparse dataset. This, in turn, shows the relative importance of the virtual motion profile of a given content to assessing and predicting the cybersickness likelihood.

Finally, we believe that there should also be an open discussion on what and how to collect cybersickness data, which should be made public for this important research area to make significant progress in the coming future.

9 APPENDICES

Table 7: Features of previous datasets and associated prediction models. The data points refer to the number of cybersickness data points that have been collected. x indicates that the work did not use machine learning methods. (Part a)

Literature	Independent Variable	Ground Truth	Interval	Data Points	Method-Models Used	Prediction Type	Publicly Available	Provided Metrics
Jeong et al. [14]	EEG HR GSR	Binary Sickness Level (0–1)	Continuous (100–120Hz)	Hard to estimate (100–120Hz)	DNN CNN	Classification	X	Accuracy
Smyth et al. [36]	EDA Skin Temp	FMS	1 minute	Approx. 300 (simulator)	X	X	X	p-value Coefficient
Xue et al. [44]	Emotion EDA BVP Skin Temp	SSQ	5–6 minutes	96	Random Forest 1D CNN LSTM	Classification	O	P value
Kim et al. [21]	Video GSR HR	SSQ	1 minute	1,600	CNN+LSTM	Regression	X	PLCC SROCC RMSE
Dong et al. [6]	Timestamp Rotaion HR	SSQ	3-5 minute	150	X	X	X	–
Weina et al. [15]	Head Movement View from each eye	Ranking-rating (subjective)	3 minute	360	CNN LSTM- RNN SVR	Regression	X	R ² MSE
Lee et al. [25]	EEG EKG GSR Content View	SSQ	90 seconds	40	LSTM CNN	Regression	X	PLCC SROCC RMSE
Kim et al. [23]	EEG	Subjective(1-5)	Unknown	8,888	CNN RNN	Classification	X	Accuracy Confusion Matrix
Islam et al. [10]	Eye-tracking, Head-tracking, Stereo Images	FMS	30 seconds	1,890	CNN- LSTM	Classification Regression	X	Accuracy Precision Recall R ² PLCC RMSE
Wang et al. [42]	Eye-tracking, Head-tracking, Character motion	Binary Sickness Level(0-1)	Continuous (90Hz)	Hard to estimate	LSTM	Classification	X	Accuracy

Table 7: Collected data and its methods from previous works (Part b)

Literature	Independent Variable	Ground Truth	Interval	Data Points	Method-Models Used	Prediction Type	Publicly Available	Provided Metrics
Islam et al. [11]	Eye-tracking, Head-tracking, HR, GSR	FMS	30 seconds	1,890	LSTM, CNN	Regression	X	MAE, MSLE, sMAPE, RMSE
Jeong et al. [13]	Eye-tracking, Head-tracking, EDA, BVP, Skin Temp	FMS	15 seconds	2,700	Transformer	Regression	X	MAE, RMSE
Wen et al. [43]	Camera Movement, Object Movement, GSR, HR	FMS	30 seconds	15,000	—	Classification	X (Dead Link)	Accuracy, F1 score
Padmanban et al. [33]	Video, GSR, HR	SSQ	1 minute	1,600	Decision Tree	Regression	O	Error

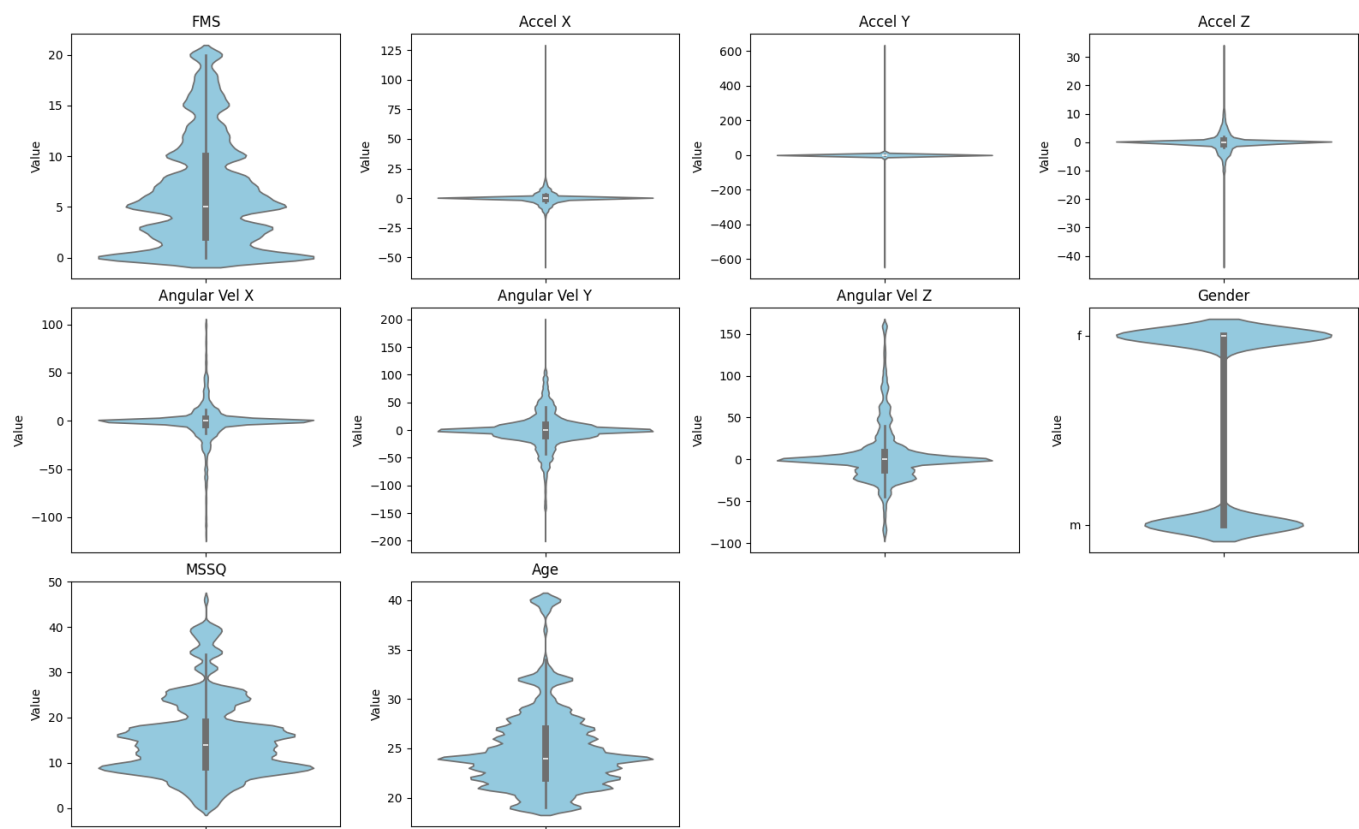


Fig. 7: Feature distribution shown as a violin plot.

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